

SAKAI, Japan

WMO: 477420

Lat: 35.5447N

Lon: 133.2350E

Elev: 4

StdP: 101.28

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.8	0.1	-7.0	2.1	3.4	-5.7	2.3	3.9	8.0	3.4	7.2	3.6	2.6	250	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	33.6	25.6	32.4	25.4	31.3	25.1	26.5	31.2	26.0	30.6	25.6	29.9	3.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.4	28.7	24.8	19.8	28.5	24.3	19.3	28.2	82.8	31.4	80.7	30.7	78.9	30.0	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	6.0	5.3	DB	-3.1	35.8	1.6	1.0	-4.3	36.5	-5.3	37.1	-6.2	37.7	-7.4	38.4	(4)
(5)			WB	-4.0	27.3	1.6	0.5	-5.2	27.7	-6.1	28.0	-7.0	28.3	-8.1	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	4.8	5.3	8.4	13.4	18.4	22.0	26.5	27.8	23.5	18.0	12.5	7.4	(6)	
	DBStd	8.30	2.33	2.66	3.08	3.24	2.61	2.35	2.60	2.18	2.64	2.65	2.91	2.72	(7)	
	HDD10.0	469	161	134	69	7	0	0	0	0	0	0	9	88	(8)	
	HDD18.3	1829	419	365	308	152	32	1	0	0	1	37	176	339	(9)	
	CDD10.0	2559	1	2	19	108	259	362	511	553	404	249	83	7	(10)	
	CDD18.3	877	0	0	0	3	33	112	252	294	154	28	1	0	(11)	
	CDH23.3	7183	0	0	1	22	184	518	2329	3132	919	77	1	0	(12)	
	CDH26.7	2420	0	0	0	2	24	99	846	1226	212	11	0	0	(13)	
(14) Wind	WSAvg	2.2	2.4	2.3	2.4	2.3	2.2	2.0	2.1	2.1	2.0	1.9	2.0	2.3	(14)	
(15) Precipitation	PrecAvg	1840	174	124	131	112	134	164	223	135	203	127	144	168	(15)	
	PrecMax	2319	263	181	200	190	354	364	589	316	420	275	229	278	(16)	
	PrecMin	1534	107	78	61	37	51	40	9	29	52	50	45	63	(17)	
	PrecStd	225	47	32	36	45	72	89	149	75	109	67	51	57	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	17.8	21.3	25.9	29.2	31.2	34.6	35.6	32.8	27.9	22.8	16.8	(19)	
		MCWB	10.4	12.8	13.0	16.7	18.5	23.2	25.6	25.8	25.1	21.6	16.7	12.6	(20)	
	2%	DB	11.5	13.6	18.2	23.3	26.9	29.2	33.2	34.1	30.9	25.3	20.3	14.7	(21)	
		MCWB	7.9	9.6	11.6	15.1	17.6	22.2	25.5	25.7	24.5	20.0	15.6	10.8	(22)	
	5%	DB	10.1	11.6	15.9	21.3	25.4	27.6	32.1	33.0	29.0	23.7	18.8	13.3	(23)	
		MCWB	6.8	8.3	11.1	14.4	17.2	21.5	25.3	25.6	23.7	18.9	14.7	9.7	(24)	
	10%	DB	8.9	10.1	14.0	19.4	23.8	26.2	31.0	31.8	27.5	22.4	17.5	12.0	(25)	
		MCWB	5.9	7.0	10.0	13.6	16.8	21.0	25.0	25.5	23.0	18.0	13.9	8.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	13.3	14.6	19.0	21.2	24.9	26.9	27.2	26.2	23.1	18.4	14.1	(27)	
		MCDB	13.4	16.7	18.6	22.1	24.8	29.2	31.8	32.0	30.4	26.1	20.6	15.6	(28)	
	2%	WB	8.8	10.5	13.0	17.0	20.0	23.8	26.3	26.7	25.2	21.4	16.9	11.7	(29)	
		MCDB	10.9	12.5	16.2	21.5	23.4	27.1	31.0	31.4	29.0	24.1	19.2	13.8	(30)	
	5%	WB	7.3	8.8	11.6	15.5	18.9	22.9	25.8	26.3	24.5	20.1	15.6	10.3	(31)	
		MCDB	9.4	11.1	14.7	19.6	22.9	25.8	30.4	30.8	28.0	22.5	18.1	12.6	(32)	
	10%	WB	6.2	7.3	10.4	14.5	18.0	22.0	25.4	25.9	23.9	18.8	14.3	9.1	(33)	
		MCDB	8.6	9.7	13.7	18.1	22.2	24.9	29.7	30.3	26.9	21.4	17.0	11.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.7	8.2	9.1	8.8	6.8	6.4	6.7	6.3	7.4	7.4	6.4	(35)	
		MCDBR	8.5	10.7	12.3	12.7	12.0	9.0	8.3	8.3	8.0	9.0	9.7	9.1	(36)	
	5% WB	MCWBR	5.8	7.3	7.5	6.1	5.1	3.7	2.9	2.9	3.3	4.5	5.6	5.9	(37)	
		MCWBR	8.0	9.2	10.2	9.8	8.9	6.7	7.7	7.3	6.9	6.9	8.0	8.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.438	0.472	0.540	0.557	0.554	0.572	0.569	0.531	0.490	0.453	0.435	0.430	(40)	
		taud	2.001	1.943	1.810	1.823	1.888	1.923	1.991	2.099	2.164	2.156	2.108	2.044	(41)	
	Ebn at Noon	Ebn at Noon	717	745	730	746	755	740	739	759	770	760	721	697	(42)	
		Edh at Noon	137	162	201	209	198	191	178	157	140	129	122	124	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.69	2.43	3.59	4.76	5.25	4.92	4.86	5.05	3.91	3.18	2.16	1.53	(44)	
	RadStd	RadStd	0.14	0.24	0.32	0.39	0.50	0.38	0.72	0.57	0.34	0.28	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page